

PITCH-CLASS STEERING FOR DIFFUSION-BASED MUSIC GENERATION VIA LATENT-SPACE PROBES

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ABSTRACT

Recent work on controllable music generation has focused on autoregressive models, leaving diffusion-based systems comparatively underexplored. We present a lightweight method for steering the pitch content of audio produced by Stable Audio Open, a latent diffusion model for music synthesis. A small convolutional probe containing approximately 125k parameters is trained to decode frame-level pitch-class activations from the model’s variational autoencoder latent space, using paired audio and MIDI data. At inference time, the frozen probe serves as a differentiable loss function: its gradient with respect to the denoising latent is used to nudge generation toward a user-specified pitch-class sequence, requiring no retraining or architectural modification of the base model. Across 27 evaluation trials spanning 9 text prompts and 3 target melodies, probe-guided generation increases melodic coherence by $2.4\times$ over the unguided baseline ($p < 10^{-5}$, Wilcoxon signed-rank test), demonstrating that musically meaningful structure is both recoverable and steerable in diffusion-based music latent spaces.

Index Terms— Controllable music generation, latent diffusion, probing, inference-time steering, pitch

1. INTRODUCTION

Text-to-music generation has advanced rapidly, with systems such as MusicGen [1], AudioLDM [2], and Stable Audio Open [3] producing audio of increasing quality and diversity. However, text prompts alone offer only coarse control over the musical content of the output. A user can request “a piano piece in C major” but cannot specify a particular melodic sequence, limiting their use in composition workflows requiring precise pitch control.

Several recent studies have begun to address this gap through inference-time intervention on model activations. Facchiano et al. [4] apply activation patching to steer binary musical attributes such as tempo and timbre in MusicGen. Singh et al. [5] and Paek et al. [6] use sparse autoencoders to discover interpretable features in autoregressive music models, while Zhao et al. [7] introduce Recursive Feature Ma-

chines for note-level control of MusicGen activations. Panda et al. [8] use linear probe weights as steering vectors for timbre and style transfer, and Jiang et al. [9] steer symbolic music generation by injecting mean activation vectors for composer style. In a complementary line of work, Koo et al. [10] probe individual attention heads in music transformers to map out what musical information different components encode.

These methods target autoregressive architectures exclusively. For diffusion-based music generation, DITTO [11] and DITTO-2 [12] optimize auxiliary losses at inference time but require per-attribute objective functions and do not leverage the latent-space structure of the underlying variational autoencoder (VAE). Music ControlNet [13] adds conditioning networks for melody and dynamics control but requires architectural modification and retraining.

We take a different approach. We first show that the VAE latent space of Stable Audio Open encodes pitch-class information in a form that a small convolutional probe can decode at frame level. We then repurpose this frozen probe as a differentiable loss function during diffusion sampling: at each denoising step within a specified window, the gradient of a binary cross-entropy loss between the probe’s pitch-class predictions and a target pitch-class sequence is computed and applied as an RMS-normalized update to the latent. The base model is never retrained; the probe, containing approximately 125k parameters, is the only learned component.

Our contributions are twofold. First, we demonstrate that frame-level pitch-class information is decodable from the Stable Audio Open VAE latent space with a micro-F1 score of 0.66, and that this decodability remains robust under diffusion noise levels encountered during the denoising process. Second, we propose probe-guided latent steering for diffusion-based music generation and show that it significantly improves melodic coherence over unguided generation across a range of guidance strengths and target melodies.

2. RELATED WORK

Controllable music generation has been approached from several directions. One family of methods modifies the generation model: Music ControlNet [13] trains additional condi-

tioning modules for melody and dynamics control of diffusion models, and Mustango [14] conditions on musical attributes through text augmentation. These approaches require retraining or architectural changes. A second family operates at inference time: DITTO [11] and DITTO-2 [12] optimize differentiable losses during the denoising process of a music diffusion model but require hand-designed loss functions for each attribute and do not exploit latent-space geometry. In the autoregressive setting, Facchiano et al. [4] apply activation patching for attribute control in MusicGen, Panda et al. [8] repurpose probe weights as steering vectors, Zhao et al. [7] apply Recursive Feature Machines for note-level control, and Jiang et al. [9] inject mean activation vectors for composer style. Our method bridges these directions: like DITTO it operates at inference time on a diffusion model, but like the autoregressive methods it derives its control signal from a learned probe.

Probes have been widely used to assess what information neural representations encode [15]. Wei et al. [16] probe MusicGen for music-theoretic concepts, Koo et al. [10] probe attention heads in music transformers, and Singh et al. [5] and Paek et al. [6] train sparse autoencoders on music model activations to discover interpretable features. Our work extends probing from a diagnostic tool to an active control mechanism: the probe both reveals what the latent space encodes and provides the gradient signal for steering.

Huang et al. [17] systematically study evaluation metrics for text-to-music systems and find that existing automated metrics correlate poorly with human judgment. Following this insight, we use a task-specific melodic coherence metric rather than general-purpose audio quality scores.

3. METHOD

Our approach consists of two stages: training a pitch-class probe on the VAE latent space (Section 3.1) and using the trained probe to guide the diffusion sampling process at inference time (Section 3.2). We begin with a brief overview of the base model.

Stable Audio Open [3] is a text-to-music latent diffusion model. It consists of three components: an Oobleck VAE that compresses waveforms into a continuous latent space with 64 channels, a text encoder (CLAP [18]) that produces conditioning embeddings from text prompts, and a diffusion transformer that generates latent sequences conditioned on both text and timing embeddings. The VAE operates at a high compression ratio, mapping 5 seconds of 44.1 kHz audio (220,500 samples) to a latent tensor of shape $64 \times T$ where $T \approx 215$ frames. This compressed representation must retain enough musical information for the diffusion model to reconstruct coherent audio, making it a informative target for probing: if pitch information survives the compression, it can potentially be read out and used as a control signal.

3.1. Pitch-Class Probe

We train a lightweight convolutional probe to predict frame-level pitch-class activations from the VAE latent representation of Stable Audio Open. The input to the probe is a latent tensor $\mathbf{z} \in \mathbb{R}^{C \times T}$ with $C=64$ channels and T time frames produced by the Oobleck VAE encoder. The output is a multi-label prediction $\hat{\mathbf{y}} \in [0, 1]^{T \times 12}$ indicating the presence of each of the 12 pitch classes (C, C $\sharp$, D, ..., B) at each frame.

The probe architecture is a two-layer one-dimensional convolutional network. Each layer applies a 1D convolution with kernel size 5 and same-padding, followed by group normalization with 8 groups and a ReLU nonlinearity. A final 1×1 convolution projects the hidden representation to 12 output channels, and a sigmoid activation produces per-class probabilities. The hidden dimension is 128, yielding approximately 125k trainable parameters. This architecture captures short-range temporal dependencies while remaining small enough to provide a meaningful test of what the latent space encodes.

We train the probe on a portion of the MAESTRO dataset [19], which provides time-aligned audio and MIDI pairs of piano performances. Audio clips of 5 seconds are encoded through the frozen VAE to produce latent representations, and pitch-class labels are extracted from the aligned MIDI files using the `pretty_midi` library [20]. After splitting at the file level to prevent data leakage from clips originating in the same recording, we obtain 2,637 training clips and 417 validation clips.

Because the probe will be applied to partially denoised latents during inference, we augment the training data with additive Gaussian noise at scales $\sigma \in \{0, 0.1, 0.3, 0.5, 0.7\}$, sampled uniformly per batch element. The noised latent is then RMS-normalized to preserve the original magnitude scale. This augmentation ensures the probe generalizes to the range of noise levels encountered at different stages of the diffusion denoising process.

The probe is trained with binary cross-entropy loss using the AdamW optimizer with a learning rate of 10^{-3} and weight decay of 10^{-5} , cosine learning rate annealing, and gradient clipping at norm 1.0, for a total of 50 epochs.

3.2. Probe-Guided Inference

Given a text prompt and a target pitch-class sequence, we steer the standard diffusion sampling process of Stable Audio Open by using the trained probe as a differentiable loss function.

The user provides a list of MIDI note numbers and a tempo, which are converted to a frame-level pitch-class target $\mathbf{y}^* \in \{0, 1\}^{T \times 12}$ by mapping each note to its pitch class and distributing note onsets across frames according to the specified tempo and note duration. This representation abstracts away octave information, focusing the guidance signal on pitch-class content rather than absolute register.

Stable Audio Open uses a denoising schedule of $N=50$ steps with classifier-free guidance at scale 4.0. We apply probe guidance starting at step $\lfloor 0.4N \rfloor$ (after 40% of denoising is complete) and continuing to the final step, intervening every 2 steps. Guidance is not applied during the initial 40% of denoising because the latent at early steps is dominated by noise and the probe’s predictions are unreliable; allowing the diffusion model to establish coarse structure before steering begins produces better results than intervening from the start.

At each guidance step, the current latent $\mathbf{z}_t$ is cloned and passed through the probe in float32 precision to obtain pitch-class logits $\hat{\mathbf{y}} = f_\theta(\mathbf{z}_t)$. The binary cross-entropy loss $\mathcal{L} = \text{BCE}(\hat{\mathbf{y}}, \mathbf{y}^*)$ is computed between the probe predictions and the target, and the gradient $\mathbf{g} = \nabla_{\mathbf{z}_t} \mathcal{L}$ is obtained via back-propagation. The latent is then updated as

$$\mathbf{z}_t \leftarrow \mathbf{z}_t - \text{clamp}\left(\lambda \frac{\text{rms}(\mathbf{z}_t)}{\text{rms}(\mathbf{g})} \mathbf{g}, \pm \alpha \text{rms}(\mathbf{z}_t)\right), \quad (1)$$

where λ is the guidance scale and $\alpha=0.05$ is the maximum update ratio.

3.3. Theoretical Analysis

We motivate the update rule in Equation 1 by connecting it to classifier guidance [21] and analyzing its stability properties.

In standard classifier guidance, the score function of a conditional diffusion model is decomposed as

$$\nabla_{\mathbf{z}} \log p(\mathbf{z}_t | y) = \nabla_{\mathbf{z}} \log p(\mathbf{z}_t) + \gamma \nabla_{\mathbf{z}} \log p(y | \mathbf{z}_t), \quad (2)$$

where $p(y | \mathbf{z}_t)$ is a noise-conditional classifier and γ controls the guidance strength [22]. Our setting differs from this formulation in one important respect: the probe f_θ is trained on VAE-encoded clean latents augmented with noise, but is not conditioned on the denoising step t . The negative gradient of the BCE loss is therefore best understood as a step-agnostic approximation to $\nabla_{\mathbf{z}} \log p(\mathbf{y}^* | \mathbf{z}_t)$ rather than a strict implementation of classifier guidance. Empirically, the noise augmentation during probe training (Section 3.1) and the stability properties analyzed below are sufficient to make this approximation effective in practice.

A key challenge in applying classifier guidance to latent diffusion models is that the gradient magnitude $\|\mathbf{g}\|$ varies by orders of magnitude across denoising steps due to the changing signal-to-noise ratio of $\mathbf{z}_t$. Without normalization, a fixed step size γ would produce negligible updates at some steps and destructive ones at others. We address this through RMS normalization. Define the unnormalized update direction $\delta = \lambda \frac{\text{rms}(\mathbf{z}_t)}{\text{rms}(\mathbf{g})} \mathbf{g}$. The RMS of this update satisfies

$$\text{rms}(\delta) = \lambda \cdot \text{rms}(\mathbf{z}_t), \quad (3)$$

which follows directly from the definition of $\text{rms}(\cdot)$ and the scalar scaling. This ensures that the typical magnitude of each

element in the update is a fixed fraction λ of the typical magnitude of the latent, regardless of the gradient’s absolute scale. The update is therefore automatically adaptive: as $\text{rms}(\mathbf{z}_t)$ decreases during denoising (the signal becomes cleaner), the update magnitude decreases proportionally, applying progressively gentler corrections in later steps where the latent carries more fine-grained structure.

The element-wise clamp at $\pm \alpha \cdot \text{rms}(\mathbf{z}_t)$ provides a worst-case bound. For any element i of the updated latent $\mathbf{z}'_t$, we have

$$|z'_{t,i} - z_{t,i}| \leq \alpha \cdot \text{rms}(\mathbf{z}_t), \quad (4)$$

bounding the maximum per-element perturbation to a fraction α of the latent’s overall scale. With $\alpha=0.05$, no single guidance step can shift any latent element by more than 5% of the RMS. We found empirically that this bound prevents the mode collapse and audible artifacts observed at larger values of α , while still permitting cumulative perturbation across the guidance window to meaningfully shift the generated pitch content.

4. EXPERIMENTAL SETUP

We evaluate probe-guided steering on a set of 27 trials constructed from 9 text prompts crossed with 3 target melodies. The text prompts describe piano music with varying stylistic characteristics, ranging from classical and jazz to ambient and cinematic styles. The three target melodies are: an ascending diatonic scale from F3 to F4 (8 notes), a descending scale from F4 to F3 (8 notes), and an alternating arpeggio pattern (7 notes). All melodies are specified in F major and use a tempo of 120 BPM with one note per beat, yielding target durations of 3.5 to 4 seconds within the 5-second generation window. Generated audio is 5 seconds long at a sampling rate of 44.1 kHz.

We compare four probe guidance scales ($\lambda=0.03, 0.05, 0.10$, and 0.20) against an unguided baseline that uses the same prompts and random seed but applies no probe intervention. All conditions share the same diffusion hyperparameters: 50 denoising steps, classifier-free guidance scale of 4.0, and a fixed random seed.

To quantify the degree of pitch-class control, we define a melodic coherence metric. We first extract fundamental frequency contours from the generated audio using the pYIN algorithm [23] as implemented in librosa [24]. Detected pitches are converted to pitch classes and compared frame-by-frame against the target pitch-class schedule within regions where the target specifies an active note. Melodic coherence is defined as the fraction of voiced frames in target-active regions where the detected pitch class matches the target; throughout this paper we use this term as shorthand for this frame-level pitch-class match rate, and it does not capture higher-order melodic properties such as interval structure or rhythmic alignment. Trials in which fewer than 20% of target-

Table 1. Probe decoding performance under additive Gaussian noise. The probe maintains usable accuracy across noise levels encountered during diffusion denoising.

Noise σ	F1 (micro)	Positive recall
0.0	0.663	0.596
0.1	0.662	0.595
0.3	0.657	0.588
0.5	0.642	0.569
0.7	0.621	0.542
1.0	0.579	0.493

region frames are voiced are excluded to avoid unreliable estimates; in practice, no trials were excluded.

Statistical significance is assessed using the Wilcoxon signed-rank test (one-sided) on paired per-trial coherence differences between each guided method and the baseline, computed across all 27 trials.

5. RESULTS

5.1. Probe Decoding Performance

The convolutional probe achieves a micro-F1 score of 0.663 and a macro-F1 of 0.657 on held-out validation data, with per-class F1 scores ranging from 0.620 to 0.683 across all 12 pitch classes. The positive-class recall, measured as the fraction of frames containing at least one active pitch class where the probe correctly identifies at least one, is 0.596. These results confirm that the Stable Audio Open VAE latent space encodes pitch-class information in a form accessible to a small nonlinear probe. The relatively uniform per-class performance suggests that no particular pitch class is privileged or suppressed in the latent representation, which is important for unbiased steering across arbitrary target melodies.

Table 1 examines how probe performance degrades as Gaussian noise is added to the latent representations, simulating the conditions encountered during diffusion denoising. Performance decreases gradually from an F1 of 0.663 at $\sigma=0$ to 0.579 at $\sigma=1.0$, indicating that pitch-class information remains substantially accessible even at noise levels well beyond those used during training augmentation. This robustness is essential for the guidance scheme, since the probe must produce a useful gradient signal when applied to partially denoised latents that still contain significant noise.

5.2. Melodic Coherence

Table 2 and Figure 1 summarize the main results. All four probe-guided conditions significantly outperform the unguided baseline, increasing mean melodic coherence from 0.112 to a range of 0.236 to 0.274 depending on the guidance scale. The best-performing condition ($\lambda=0.05$) achieves a

Table 2. Melodic coherence across probe guidance scales. All guided conditions significantly outperform the unguided baseline (Wilcoxon signed-rank test, $n=27$ paired trials).

Method	Mean $\pm$ Std	Median	p -value
Baseline	0.112 ± 0.104	0.098	—
Probe ($\lambda=0.03$)	0.236 ± 0.104	0.244	4.6×10^{-4}
Probe ($\lambda=0.05$)	0.274 ± 0.084	0.269	5.7×10^{-6}
Probe ($\lambda=0.10$)	0.266 ± 0.071	0.263	8.0×10^{-6}
Probe ($\lambda=0.20$)	0.273 ± 0.064	0.262	2.3×10^{-6}

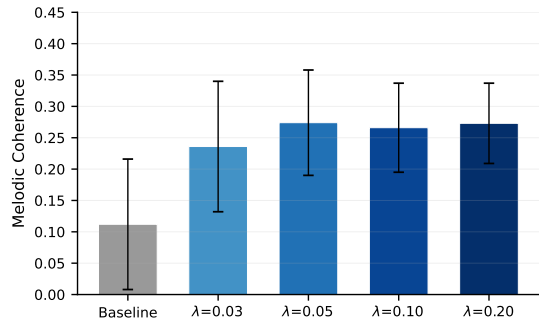


Fig. 1. Mean melodic coherence with standard deviation error bars across guidance scales. The baseline (gray) uses no probe intervention. All probe-guided conditions (blue) substantially improve pitch-class alignment.

$2.4\times$ improvement in mean coherence over the baseline, and all guided conditions yield highly significant improvements ($p < 5 \times 10^{-4}$).

Coherence improves rapidly from $\lambda=0.03$ to $\lambda=0.05$ and then plateaus for stronger guidance scales. This saturation suggests that a moderate guidance strength is sufficient to exploit the available pitch-class signal in the latent space, and that increasing the scale further does not yield additional pitch accuracy. At the same time, the standard deviation decreases monotonically with increasing guidance strength, from 0.104 at $\lambda=0.03$ to 0.064 at $\lambda=0.20$, indicating that stronger guidance produces more consistent steering across diverse prompts and melody types even when the mean improvement has leveled off.

Table 3 breaks down the results by melody type at the best operating point ($\lambda=0.05$). All three target patterns show substantial improvements over the baseline, confirming that the steering mechanism generalizes across different pitch-class trajectories. The alternating arpeggio achieves the highest guided coherence, possibly because its wider pitch-class intervals create a more distinctive target signal for the probe gradient.

Table 3. Per-melody coherence breakdown at $\lambda=0.05$ ($n=9$ prompts per melody type). All melody types show substantial improvement.

Melody	Baseline	Probe ($\lambda=0.05$)
Ascending	0.095 ± 0.081	0.268 ± 0.074
Descending	0.144 ± 0.108	0.234 ± 0.064
Alternating	0.096 ± 0.112	0.320 ± 0.090

6. DISCUSSION

The results demonstrate that the VAE latent space of a diffusion-based music model encodes pitch-class information that is not only decodable but also actionable for generation steering. The probe’s moderate F1 of 0.66 might suggest limited utility as a control signal, yet the downstream coherence improvements are large and statistically robust. This can be understood by noting that steering does not require perfect frame-level predictions; rather, it requires that the gradient of the probe loss point in a direction that increases the likelihood of the target pitch content in the decoded audio. An imperfect probe can still provide a useful directional signal because the gradient aggregates information across all pitch classes and time frames simultaneously.

The adaptive behavior predicted by the analysis in Section 3.3 is confirmed empirically: RMS normalization produces naturally gentler corrections in later denoising steps where the latent carries more fine-grained structure, without requiring a manually tuned step-size schedule. The saturation of coherence beyond $\lambda=0.05$, combined with decreasing variance at higher scales, suggests that stronger guidance primarily reduces outlier trials rather than improving the typical case, a desirable property for practical use.

All experiments use the MAESTRO piano dataset for probe training while evaluating on diverse piano-related text prompts, demonstrating domain transfer. This transfer is likely facilitated by the fact that pitch-class structure is a low-level musical property that generalizes across piano styles. Our evaluation is limited to a single base model (Stable Audio Open), pitch-class control without full melodic specification including rhythm and register, and simple diatonic test melodies. We leave generalization to other diffusion models, richer musical targets, and broader timbral evaluation to future work.

7. CONCLUSION

We have presented a lightweight approach to pitch-class steering in diffusion-based music generation. By training a small convolutional probe on the VAE latent space of Stable Audio Open and repurposing it as a differentiable loss at inference time, we achieve significant and consistent improvements in

melodic coherence without retraining the base model. The method requires only approximately 125k additional parameters, adds minimal computational overhead, and accommodates arbitrary pitch-class targets at inference time.

More broadly, our results suggest that latent diffusion models for music encode musically meaningful structure in their intermediate representations, and that lightweight probes offer a practical means of both analyzing and exploiting this structure for controllable generation. The probe-as-loss paradigm is not specific to pitch: the same framework could in principle be applied to any musical attribute decodable from the VAE latent space, including chord quality, rhythmic density, or timbral characteristics, by training an appropriate probe and using its gradient as the steering signal.

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